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RESULTS

Results

01 · NESTED ANOVA

one factor nested within another

Table 1. Descriptives by top-level group

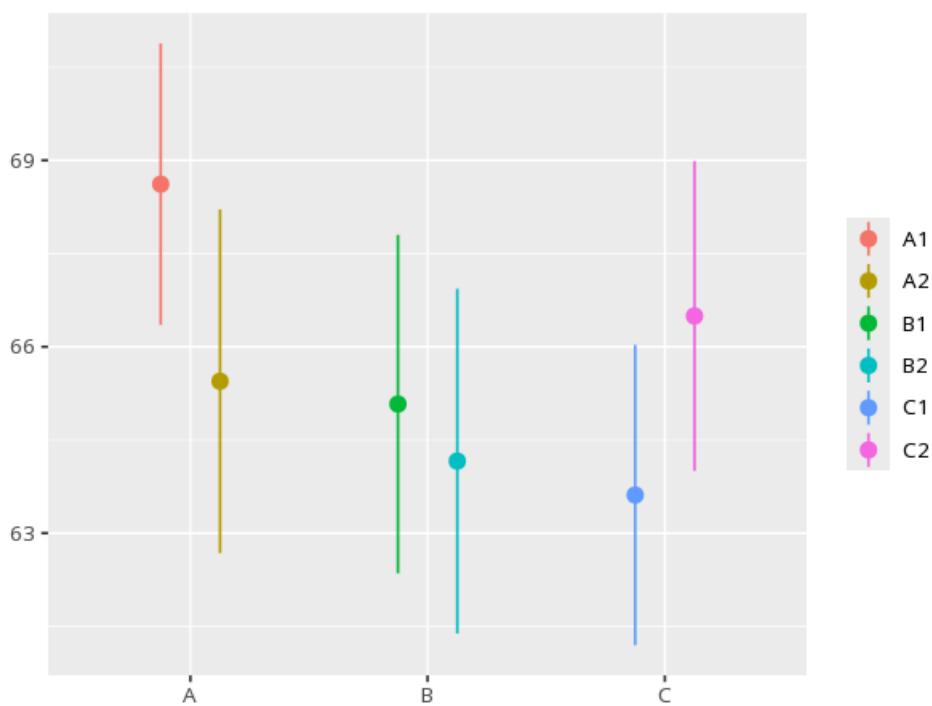
Group	N	Mean	Std. Dev.
A	80	67.03	8.01
B	80	64.62	8.55
C	80	65.05	7.77

Table. Nested ANOVA

Source	SS	df	MS	F	p	ω^2 [95% CI]
school	264.38	2	132.19	1.03	.456	0.01 [0.00, 1.00]
classroom (nested in school)	384.23	3	128.08	1.97	.120	0.01 [0.00, 1.00]

Under random nesting the F for the upper factor (A) uses the nested factor's mean square B(A) as its error term, while B(A) is tested against the residual — so the two F rows do not share the same denominator. Variance components (or ω^2) are reported as the effect size where estimable. Assumption checks: Levene's (equal variances across top-level groups) & normality of residuals (Shapiro-Wilk). (Levene F=0.43, p=.652 · Shapiro W=0.99, p=.381) — equal variances look reasonable — residual normality looks reasonable

Figure. Grouped means



Understanding the numbers

Source. Names which term the row tests: the top-level factor (A), or the nested factor (B) within it. Here, the headline row is school.

SS. The variability in the outcome attributable to that term. Here, $SS = 264.38$.

df. The degrees of freedom for that term. Here, $df = 2$.

MS. That term's SS divided by its df. Here, $MS = 132.19$.

F. The ratio of that term's mean square to its own error term - the top factor is tested against the nested factor's variability, not raw residuals. Here, $F = 1.03$.

p. The probability of an F this large if that term truly had no effect. Here, $p = .456$ - at or above the $\alpha = 0.05$ threshold.

ω^2 . The proportion of variance explained by the top-level factor, a less-biased alternative to η^2 ($\sim .01$ small, $.06$ medium, $.14$ large). Here, $\omega^2 = .01$ [.00, 1.00].

Mean. The average outcome for that top-level group. Here, each of the 3 top-level groups has its own mean in the descriptives table above.

N. The number of observations in that top-level group. Here, each of the 3 top-level groups has its own N in the descriptives table above.

Std. Dev.. How spread out outcome scores are within that top-level group. Here, each of the 3 top-level groups has its own SD in the descriptives table above.

How to read this test

Used when one factor sits inside another (e.g. classes within schools). The top factor is tested against variation among its nested units, not raw residuals — a significant F means the top-level groups differ beyond the nested-unit variability. Your significance threshold (α) is 0.05.

APA template: A nested ANOVA for A gave $F(2,3)=1.03$, $p = .456$, $\omega^2=.01$ [.00, 1.00].

Statistical basis: Nested ANOVA (upper factor tested against the nested-unit mean square) (Fisher, R. A. (1925). Statistical Methods for Research Workers. Oliver & Boyd.); ω^2 effect size (Ben-Shachar MS, Lüdtke D, Makowski D (2020). "effectsize: Estimation of Effect Size Indices and Standardized Parameters." Journal of Open Source Software, 5(56), 2815.). Full references in the exported CITATIONS.txt.

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